

MERN & Next.js Full-Stack Project Ideas for B.Tech Students

MERN (MongoDB, Express.js, React, Node.js) and Next.js are powerful JavaScript frameworks for building modern full-stack applications ¹. Each project idea below involves building a complete front-end (React or Next.js) and back-end (Node/Express) with a MongoDB database, authentication, and RESTful APIs. These projects simulate real-world applications and are suitable for deploying on GitHub as portfolio projects.

1. E-Commerce Marketplace (MERN)

Build a full-featured online store where users can browse products, add items to a cart, and complete secure checkout. Implement user accounts (customers and admins), product management, and order processing. **An e-commerce platform is one of the most prominent MERN project ideas**, demonstrating many skills in one app ².

Key features:

- User registration/login with JWT-based authentication and role-based access (customer vs admin)
- Product catalog with full CRUD operations (Express REST API, MongoDB)
- Shopping cart and checkout flow with payment gateway integration (Stripe/PayPal)
- Admin dashboard to manage inventory, orders, and view sales reports
- Responsive UI (using Material-UI or Bootstrap) and image uploads (e.g. via Cloudinary)

2. Real-Time Chat Application (MERN)

Create a messaging app that allows users to chat in real time (one-to-one and group conversations). Use Socket.IO (WebSockets) on a Node/Express server to push and receive messages instantly. Such a chat app provides hands-on practice with real-time data and event-driven architecture ³.

Key features:

- User profiles and authentication (persist sessions with JWT or cookies)
- Real-time message exchange using WebSockets (Socket.IO)
- Chat rooms or direct messaging; display online/offline status and typing indicators
- Store chat history and user data in MongoDB; implement conversation threads with timestamps
- Optional: File/image sharing, emojis, and notifications for new messages

3. Blogging Platform / CMS (Next.js)

Develop a blog content-management platform where authenticated authors can create, edit, and publish posts. Leverage Next.js for server-side rendering (SSR) or static generation of public blog pages for better SEO. Include features like rich-text editing (e.g. Markdown) and comments. Users can browse recent posts and leave comments if logged in ⁴.

Key features:

- User authentication (authors and readers) with secure sign-up/sign-in (NextAuth or custom JWT)
- Admin interface for writing and editing articles (Markdown or WYSIWYG editor)
- RESTful API routes (Next.js API routes or Express) to handle CRUD for posts and comments

- Comment section under each post with nesting; spam-moderation tools for admins
- SEO-friendly pages using Next.js SSR/SSG; responsive design for mobile reading

4. Kanban Project Management Tool (MERN)

Implement a Trello-like Kanban board app for managing tasks and projects. Users can create multiple boards with lists (columns) and cards (tasks), then drag-and-drop cards between lists. This project exercises complex front-end interactions and persistent state via an API. You can also add real-time updates with WebSockets to sync board changes across clients.

Key features:

- Boards with customizable lists and task cards; drag-and-drop UI using a library (e.g. React DnD)
- User authentication and team collaboration (invite users to boards)
- REST API (Express/MongoDB) to create, update, and delete boards, lists, and tasks
- Real-time sync (Socket.IO) so that multiple users see board changes live
- Task details with due dates, attachments, and labels; progress indicators (e.g. percent complete)

5. Social Networking / Q&A Platform (MERN)

Build a community-driven site, like a microblogging/social network or a Q&A forum (similar to StackOverflow). Users have profiles and can post content (text, links, images). Other users can upvote, comment, or answer questions. This covers user relationships (following), content feeds, and interactive features.

Key features:

- User profiles, follow/followers system, and personalized feeds of posts or questions
- Authenticated posting: create and edit posts or questions with rich content
- Upvotes/likes and comments on posts; reputation or scoring system for activity
- REST API for posts, comments, and user interactions (using Node/Express and MongoDB)
- Real-time notifications (e.g. sockets or polling) for new replies or likes, and moderation tools

6. Online Learning Platform (MERN/Next.js)

Create a mini Learning Management System (LMS) where instructors can upload courses and students can enroll and track progress. Features include video lectures, quizzes, and progress dashboards. Next.js can be used to render public course pages, while a MERN back-end handles user data, content, and REST APIs.

Key features:

- Course catalog: list courses with details, and allow users to enroll
- Video content delivery (embedded streaming) and quizzes with immediate feedback
- Role-based access: instructor vs student dashboards with different privileges
- Progress tracking for each user (e.g. mark lessons complete, view grades)
- Certificate generation on completion, and RESTful endpoints for course content and results

7. Event Booking / Ticketing System (Next.js)

Build an application for creating and managing events and ticket sales. Event organizers can post upcoming events, and users can browse events and book tickets. Use Next.js for server-rendered event pages and integrate a REST API for bookings. Include features like seat selection, order history, and email confirmations.

Key features:

- Event listings with details (date/time, location, description) and search/filter function
- User auth for attendees and organizers; organizers can create and manage their events
- Ticket booking system with seat selection or ticket quantity and payment integration (Stripe/PayPal)
- User dashboard showing booked events and downloadable e-tickets (e.g. QR code)
- Calendar integration (add events to Google Calendar) and email notifications on booking

8. Collaborative Whiteboard / Workspace (MERN)

Develop a shared drawing board or workspace where multiple users can collaborate in real time. This app simulates remote team collaboration tools. Users can draw on a canvas, add text or shapes, and see others' updates instantly (using Socket.IO or WebRTC). Store completed sessions so users can save or review whiteboards later.

Key features:

- Online canvas with drawing tools (pen, shapes, text) and undo/redo functionality
- Real-time collaboration: sync board state via WebSockets so all participants see changes
- Session management: create or join rooms for separate whiteboards; user cursors with names
- Save boards to MongoDB and allow exporting (PNG or PDF)
- Chat or voice integration (WebRTC) as an optional extra feature for teams

9. Photo-Sharing Social App (MERN)

Create an Instagram-like photo-sharing platform. Users can upload photos (with captions) and follow each other. Build a feed of images from followed users, and allow liking and commenting on photos. This project emphasizes media handling, user-generated content, and social interactions.

Key features:

- User profiles with avatar and bio; image upload (store on AWS S3 or Cloudinary) and carousel/gallery
- Follow system: users see a personalized feed of photos from people they follow
- Likes and comments on photos; real-time updates or refreshing feed
- Search by tags or location and trending hashtags
- Notifications for new followers, likes, or comments (e.g. via background jobs or WebSockets)

10. Fitness/Health Tracker App (MERN)

Build an application that lets users track workouts, diet, or health goals over time. Each user has a dashboard showing their logged activities and progress charts. **This type of app showcases data visualization and user privacy**, like a MERN fitness-tracker project which emphasizes high user engagement and polished UI/UX ⁵. You can include social features like sharing achievements or a community leaderboard.

Key features:

- User accounts with secure login; profile data (age, height, weight, fitness goals)
- Logging module for workouts or meals (type, duration, calories) with timestamps
- Visualize data with charts/graphs (using Chart.js or D3) for weekly/monthly trends
- Set and edit goals (e.g. weight target) and track completion percentage
- Optional: Peer comparisons or community challenges (team competitions)

Sources: MERN stack allows rapid full-stack development ¹. These project ideas are inspired by industry-standard examples (e.g. e-commerce, blogging, chat apps) to demonstrate practical skills ²

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1 2 3 4 5 35+ MERN Stack Projects That Impress Recruiters (2025)

<https://www.upgrad.com/blog/mern-stack-project-ideas/>